

BIBEK SAMANTA (He/Him)

International Graduate School BACCARA
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RESEARCH INTEREST

My research focuses on understanding the structure-ion transport relationships in sodium- and lithium-based solid electrolytes, with an emphasis on inorganic systems, such as argyrodites, (oxy)sulfides, halides, and oxyhalides. I specialize in multinuclear solid-state NMR techniques to investigate local structures and ion-dynamics. I am particularly interested in exploring novel inorganic Na/Li-ion conductors through both established strategies (*e.g.*, iso-/aliovalent cation or anion substitution) and emerging approaches like dual-anion chemistry. Additionally, I am interested in studying the influence of synthesis methods on ion transport properties.

DOCTORTAL THESIS

(Contract: August 2022 - July 2025)

Topic: Elucidating Structure and Ion-dynamics in Inorganic Na⁺- and Li⁺-Solid Electrolytes using Solid-State NMR

Supervisors: Prof. Dr. Michael Ryan Hansen and Prof. Dr. Wolfgang G. Zeier

Aim:

- Analysis of local structure (cation/anion coordination environments) using solid-state magic-angle spinning (MAS) NMR
- Study of cation dynamics (kHz-MHz frequency) using motional narrowing, lineshape analysis, and spin-lattice (T₁) relaxometry

Experimental Skill:

- Synthesis: sol-gel, ball mill, and hydrothermal procedure
- Analytical techniques and interpretation of results: NMR, XRD, EIS, TEM, SEM
- Investigation of different class of materials: Na-sulfides, -metal chlorides, -oxyhalides; Li-argyrodites, -halides, -sulfides
- NMR rotor packing inside glovebox (rotor size from 7 mm to 2.5 mm) for air/moisture sensitive samples
- Single pulse MAS NMR experiments at variable temperatures (nuclei: ⁶Li, ⁷Li, ²³Na, ³¹P)
- Variable temperature static experiments (single pulse, saturation recovery) for observing cation dynamics (nuclei: ⁷Li, ²³Na, ³¹P)
- Data analysis using Bruker TopSpin 4.1.1, ssNake v1.4 and Origin software
- Learning Na-based solid-state cell preparation

EDUCATIONAL QUALIFICATIONS

Degree	University/Board	Institution	Year	CGPA/%	Grade
MTech	IIT Bombay	IIT Bombay	2020-2022	9.82/10	AA
MSc	Calcutta University	Ramakrishna Mission Vidyamandira	2018-2020	9.52/10	O
BSc	Calcutta University	Ramakrishna Mission Vidyamandira	2015-2018	9.66/10	O
Higher Secondary	WBCHSE	Contai High School	2015	92.6%	O
Secondary	WBBSE	Kharui Union High School	2013	89.4%	A+

AWARDS & ACHIEVEMENTS

11. Selected to attend 74th Lindau Nobel Laureate Meetings in Chemistry	2025
10. Completed an NPTEL course on ‘Crystals, Symmetry and Tensors’ by Dr. Rajesh Prasad (Silver Medal)	2024
9. Awarded 3 rd place for poster competition at BACCARA Power Day 2023	2023
8. Membership of ECS Student Chapter Münster	2022
7. Selected for BACCARA fellowship for pursuing doctoral studies at University of Münster	2022
6. Secured All India Rank 20 in GATE 2020 in Engineering Sciences (XE) paper	2020
5. Silver medal (second rank) in MSc	2020
4. Stood 2 nd in the NPTEL course ‘Introduction to Materials Science and Engineering’ by Dr. Rajesh Prasad	2019
3. Selected in Indian Academy of Science - Summer Research Fellowship Program for two consecutive years	2018-2019
2. Gold medal (first rank) in BSc	2018
1. Awarded with ‘Swami Vivekananda Merit-cum-Means Scholarship’ and ‘Priyamvada Birla Scholarship’	2015-2018

CONFERENCES

(Conference participation from September 2022 onwards is funded by the International Graduate School BACCARA)

13. Double Anniversary of MEET Battery Research Center and Helmholtz Institute Münster, Germany	July 2025
12. 74th Lindau Nobel Laureate Meetings in Chemistry , Lindau, Germany	June - July 2025
11. Battery 2030+ Annual Conference , Münster, Germany (Talk and Poster)	May 2025
10. Solid-State Batteries VI , Frankfurt am Main, Germany (Poster)	November 2024
9. 59 th Scientific Meeting of the Dutch NMR Discussion Group, University of Twente, Netherlands (Poster)	November 2024
8. BACCARA Power Day 2024, Münster, Germany (Poster)	November 2024
7. 1st National Meeting for Sodium(-Ion) Batteries (NMSB-1) , IIT Bombay, India (Poster)	October 2024
6. 24th International Conference on Solid State Ionics (SSI24) , London, United Kingdom (Poster)	July 2024
5. 1 st #BatteryCityMünster PhD Day, Münster, Germany (Poster)	September 2023
4. BACCARA Power Day 2023, Münster, Germany (Poster, awarded 3 rd place)	September 2023
3. Summer School “Relaxation”, Windischleuba, Germany	March-April 2023
2. Sodium Battery Symposium (SBS)-3 , Berlin, Germany	September 2022
1. National Conference on Recent Developments in Nanoscience and Nanotechnology, Jadavpur, India	January 2019

ORGANIZATIONAL & SOCIAL ACTIVITIES

- Organizing committee member** for one-day conference ‘BACCARA Power Day’ in 2023, 2024 and 2025
 - Selected speakers and posters from external applicants with my co-organizers
 - Managed the official LinkedIn account of our graduate school for online promotion of the event
- Moderation** of four BACCARA Public Lectures
 - Speakers were Prof. Christian Masquelier, Prof. Matteo Bianchini, Prof. Michael Ryan Hansen and Dr. Sathiya Mariyappan
- Active participation in several socializing events** organized by both of my working groups (group retreats, PhD dinners) and our graduate school (BACCARA Bootcamp, picnics, Christmas market visit)
- Participation in several workshops** organized by our graduate school ranging from mental health awareness, improvement of presentation skills to working in multicultural teams
- Participated in MünsterXchange programs where the new doctoral students meet elder residents of Münster (who had studied at Münster University) and engage in light-hearted conversation

TEACHING EXPERIENCE

1. Teaching assistant , Physikalische Chemie Grundpraktikum, Institute of Physical Chemistry, University of Münster (30 students; conducted pre-experiment viva, corrected protocols and supervised final exams)	Summer semester 2024
2. Teaching assistant , Physikalische Chemie Grundpraktikum, Institute of Physical Chemistry, University of Münster (44 students; conducted pre-experiment viva, corrected protocols and supervised final exams)	Summer semester 2023
3. Teaching assistant , Advanced Ceramics, Metallurgical Engineering and Materials Science Department, IIT Bombay (15 students; arranged online quiz and crib sessions, proctored online exams and evaluated answer sheets for end-semester examinations)	January-March 2022
4. Teaching assistant , Advanced Composite, Metallurgical Engineering and Materials Science Department, IIT Bombay (19 students; arranged online interactive sessions, proctored online exams and evaluated answer sheets)	July-November 2021

ADDITIONAL INFORMATION

- I love calligraphy, cooking Bengali lunch and dinner recipes, swimming, travelling, reading books about my culture and country
- I regularly help students with their PhD applications and share my own experience, usually via LinkedIn
- Languages: Bengali (native), English (full proficiency), Hindi (working proficiency), German (A2)